

Practice Quiz: Learning

Part A: Multiple Choice

1. Spike-timing-dependent plasticity (STDP) implements learning by: A) Randomly strengthening or weakening synapses B) Changing synaptic strength based on the temporal relationship between pre-synaptic (prediction) and post-synaptic (input) firing C) Only strengthening synapses, never weakening them D) Operating independently of neural activity patterns
2. Bayesian Model Reduction is proposed to occur during: A) Active waking cognition B) NREM sleep — when the brain replays experiences and prunes unnecessary model complexity C) REM dreaming only D) Exercise-induced neurogenesis
3. The synaptic homeostasis hypothesis proposes that sleep: A) Globally strengthens all synapses B) Globally weakens synapses, preserving only the most informative connections C) Has no effect on synaptic strength D) Operates only on motor learning circuits
4. Critical periods are windows of heightened plasticity characterized by: A) High precision on prediction errors, driving rapid model updating B) Complete absence of prediction errors C) Reduced neural activity D) Exclusive growth of inhibitory neurons
5. PTSD in Active Inference terms involves: A) Complete loss of the generative model B) High-precision priors about danger that resist updating despite evidence of safety C) Low precision on all predictions D) Exclusive damage to the hippocampus
6. Expert performance (e.g., radiological detection in 200ms) is characterized by: A) Slower, more deliberate processing B) A simplified model with fewer features C) Deep hierarchical generative model with high-precision priors and automated policies D) Conscious analysis of each feature
7. Exposure therapy for PTSD works by: A) Erasing the traumatic memory B) Generating prediction errors (exposure to trauma cues without harm) that gradually reduce the

precision of danger priors C) Increasing avoidance behavior D) Pharmacologically blocking all emotional responses

Part B: Short Answer

1. Explain how the "stability-plasticity dilemma" (maintaining stable knowledge while remaining open to new learning) maps onto precision dynamics in Active Inference. How does the brain solve this?
2. A patient with early-stage Alzheimer's disease repeats the same stories and fails to learn new information. Analyze this in Active Inference terms: what synaptic plasticity mechanisms have been compromised?
3. Design an experiment to test whether a nap (allowing NREM replay) improves generalization performance on a newly learned category, as Bayesian Model Reduction would predict.